

TAWNY SIT

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EDUCATION

The Ohio State University Ph.D. Candidate, Astronomy	Columbus, OH August 2022 - Present
California Institute of Technology B.S. Astrophysics	Pasadena, CA June 2020

AWARDS & HONORS

NSF Graduate Research Fellowship <i>The Ohio State University</i>	August 2024 - Present
Graduate Enrichment Fellowship <i>The Ohio State University</i>	August 2023 - August 2024
Distinguished University Fellowship <i>The Ohio State University</i>	August 2022 - August 2023

PUBLICATIONS

First Author

4. **Sit, T.**, Weinberg, D. H., & Griffith, E. J., “On the Origin of Abundance Variations in the Milky Way’s High- α Plateau,” 2025, *ApJ*, 994, 53, doi:10.3847/1538-4357/ae0b5d
3. **Sit, T.**, et al., “Chemical Cartography with APOGEE: Two-process Parameters and Residual Abundances for 288,789 Stars from Data Release 17,” 2024, *ApJ*, 970, 180, doi:10.3847/1538-4357/ad4ed2.
2. **Sit, T.**, et al., “Long-rising Type II Supernovae in the Zwicky Transient Facility Census of the Local Universe,” 2023, *ApJ*, 959, 142, doi:10.3847/1538-4357/ad036f.
1. **Sit, T.** & Ness, M. K., “The Age Distribution of Stars in the Milky Way Bulge,” 2020, *ApJ*, 900, 4, doi:10.3847/1538-4357/ab9ff6.

Contributing Author

13. Dubay, L. O., et al. “A Broken Clock Is Right Twice a Day: [Ce/Mg] Is Not a Universal Chemical Clock,” submitted, arXiv: 2608.26261
12. SDSS Collaboration, et al. “The Twentieth Data Release of the Sloan Digital Sky Survey: First All-Sky BOSS Spectra, eROSITA-SDSS-V Mapper Coordinated Observations, and a Preview of the Local Volume Mapper,” submitted, arXiv: 2607.26149
11. Griffith, E. J., et al. “Untangling the Sources of Abundance Dispersion in Low-metallicity Stars. II. Neutron Capture Elements,” 2026, *ApJ*, 1001, 193, doi:10.3847/1538-4357/ae4e24
10. Sharma, Y., et al. “Twin Peaks: SN 2021uvy and SN 2022hgk in the Landscape of Double-peaked Stripped Envelope Supernovae,” 2025, *PASP*, 137, 094101, doi:10.1088/1538-3873/ae02c6
9. Schulze, S., et al., “A cosmic formation site of silicon and sulphur revealed by a new type of supernova explosion,” 2025, *Nature*, 644, 634-639, doi:10.1038/s41586-025-09375-3
8. Johnson, J. W., et al., “The Milky Way Radial Metallicity Gradient as an Equilibrium Phenomenon: Why Old Stars are Metal-Rich,” 2025, *ApJ*, 988, 8, doi:10.3847/1538-4357/addbe5

7. Griffith, E. J., et al., “Many elements matter: Detailed abundance patterns reveal star-formation and enrichment differences among Milky Way structural components,” 2025, *AJ*, 169, 280, doi:10.3847/1538-3881/ad07f
6. Das, K. K. et al., “Low-Luminosity Type IIP Supernovae from the Zwicky Transient Facility Census of the Local Universe. I: Luminosity Function, Volumetric Rate,” 2025, *PASP*, 137, 044203, arXiv:10.1088/1538-3873/adcaeb
5. Hasselquist, S., et al., “2-process Model and Residual Abundance Analysis of the Milky Way Massive Satellites,” 2024, *ApJ*, 974, 227, doi: 10.3847/1538-4357/ad70ad
4. Das, K. K., et al., “Probing Presupernova Mass Loss in Double-peaked Type Ibc Supernovae from the Zwicky Transient Facility, ” 2024, *ApJ*, 972, 91, doi:10.3847/1538-4357/ad595f
3. Anand, S., et al., “Collapsars as Sites of r-process Nucleosynthesis: Systematic Photometric Near-infrared Follow-up of Type Ic-BL Supernovae,” 2024, *ApJ*, 962, 68, doi:10.3847/1538-4357/ad11df.
2. Das, K. K., et al., “Probing the low-mass end of core-collapse supernovae using a sample of strongly-stripped Calcium-rich Type IIb Supernovae from the Zwicky Transient Facility.” 2023, *ApJ*, 959, 12, doi:10.3847/1538-4357/acfeeb.
1. Karambelkar, V., et al., “Volumetric rates of Luminous Red Novae and Intermediate Luminosity Red Transients with the Zwicky Transient Facility.” 2023, *ApJ*, 948, 137, doi:10.3847/1538-4357/acc2b9.

PRESENTATIONS

Chemical Variation Across the Milky Way with Calibrated Abundances from APOGEE

Poster - 2025 SDSS-V Collaboration Meeting

June 2025, Heidelberg, Germany

Expanded Two-process Parameters and Abundances from APOGEE DR17

Talk - DHWFEST: Dark, Hot, Warm, and Fuzzy mattEr in Space and Time

July 2024, Salt Lake City, UT

Long-rising Type II Supernovae in the Zwicky Transient Facility Census of the Local Universe

Talk - American Astronomical Society Meeting 240

June 2022, Pasadena, CA

OBSERVING

Programs

PI: Characterizing the Multi-element Abundance Dispersion of the Milky Way Disk

Large Binocular Telescope/PEPSI (49 hours total)

Semesters 2025A, 2025B, 2026A

Experience

Large Binocular Telescope: OSURC Queue Observing (Lead)

12 nights

January 2026 - Present

Large Binocular Telescope: OSURC Queue Observing (Assistant)

6 nights

February 2025 - January 2026

Palomar Observatory 200" Hale Telescope: ZTF

12 nights

June 2021 - August 2022

OUTREACH AND SERVICE

Prospective Student Recruitment

Department of Astronomy

February - April 2024, 2025, 2026

The Ohio State University

Dance + Astrophysics Consultant

Haeli Santoro - Senior Odyssey Project

Fall 2025

Columbus, OH

AstroGenMin Lunch Organizer*Department of Astronomy*

Fall 2024 - Spring 2025

The Ohio State University

LGBTQ+ Lunch Founder/Organizer*Department of Astronomy*

Spring 2024 - Spring 2025

The Ohio State University

Planning Team*Caltech Science Olympiad Club*

Fall 2017 - Spring 2020

California Institute of Technology

TEACHING AND MENTORING

Python Bootcamp Facilitator*Summer Undergraduate Research Program (SURP)*

May 2024, 2025, 2026

The Ohio State University

Academic Facilitator*Undergraduate Residential Summer Access (URSA) Program*

May - August 2023

The Ohio State University

Primary Project Mentor

Science Internship Program

June - August 2021

UC Santa Cruz (Remote)

Rise Program Tutor*Caltech Y Rise Program*

October 2017 - June 2020

California Institute of Technology

UNDERGRADUATE/POST-BACCALAUREATE RESEARCH

Assistant Research Technician - Zwicky Transient Facility*California Institute of Technology, Department of Astronomy*

June 2021 - August 2022

Mansi Kasliwal

Post-baccalaureate Research - Halo RR Lyrae in CFTLS Deep Fields*University of California, Santa Cruz (Remote)*

July 2020 - June 2021

Raja Guhathakurta

Summer Undergraduate Research Fellowship*Columbia University, Department of Astronomy*

June - August 2019

Melissa Ness

Summer Undergraduate Research Fellowship*California Institute of Technology, Department of Astronomy*

June - August 2018

Evan Kirby

Summer Undergraduate Research Fellowship*California Institute of Technology, Department of Physics*

June - August 2017

Murray Brightman, Fiona Harrison